What is role assignment in graphs?

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Introduction

Motivation

Contextualization

Graphs modeling various realities are becoming increasingly large as graphs associated with:

- Social networks;
- Knowledge;
- Artificial Intelligence;

How can we understand and extract information from colossal graphs?

- Role assignment represents the relationship between individuals who perform the same role;
- It allows a graph to be represented by a smaller graph that models its roles.

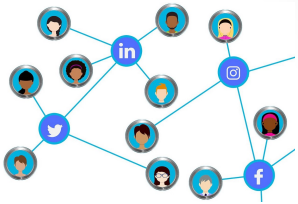


Figure 2: Social Network represented by graph.

Contextualization

Initial Contextualization

- 1991: Everett and Borgatti formalized a role assignment under the name "role coloring";
  - They defined it in terms of graph models for social networks.



Role assignment

Definition

Role assignment
Given two graphs G and R, an ***R*-role assignment** for G is a vertex mapping

$$p : V(G) \to V(R),$$

such that, for every $u \in V(G)$,

$$p(N_G(u)) = N_R(p(u)).$$

When $|V(R)| = r$, p is an ***r*-role assignment**.

We call R the role graph.

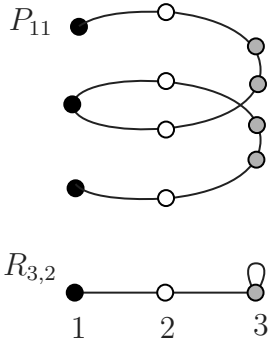


Figure 3: Graphs P_{11} e $R_{3,2}$.
P_{11} has an $R_{3,2}$-role assignment.

Decision Problems

***r*-ROLE ASSIGNMENT**
Instance: A simple graph G.
Question: Does G admit an r-role assignment?

***R*-ROLE ASSIGNMENT**
Instance: A simple graph G and a specific role graph R.
Question: Does G admit an R-role assignment?

The 2-Role Assignment Problem

Theorem 1 (Roberts and Sheng, 2001)
The 2-Role Assignment problem is NP-complete.

Theorem 2 (Roberts and Sheng, 2001)
The R-Role Assignment problem is NP-complete when $R = R_{2,5}$ or $R = R_{2,6}$.

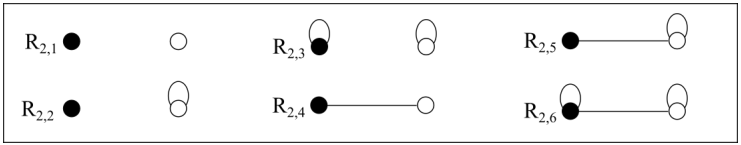


Figure 4: Possible role graphs for a 2-role assignment.

The 2-role assignment problem

Theorem 3 (Roberts and Sheng, 2001)

A graph G has an R-role assignment if and only if

R	condition
$R_{2,1}$	$E(G) = \emptyset$
$R_{2,2}$	$E(G) \neq \emptyset$ and there exist isolated vertices
$R_{2,3}$	G is disconnected with no isolated vertices
$R_{2,4}$	G is bipartite with no isolated vertices

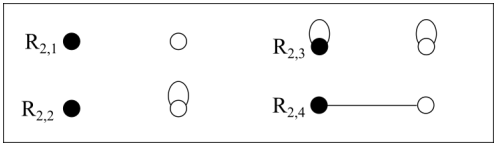


Figure 5: Role graphs from Theorem 3.

Problem of 2-role assignment - Examples

Exemple 1

The graph P_n has a 2-role assignment, but it does not have an $R_{2,6}$-role assignment.

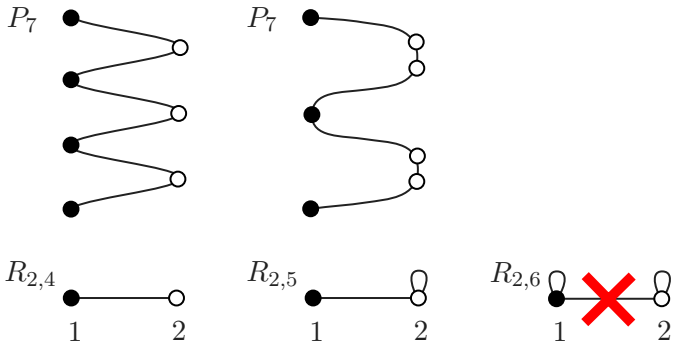


Figure 6: R-role assignments for the graph P_7 with $R = R_{2,4}$ and $R = R_{2,5}$.

2-role assignment problem - Examples

Exemple 2

The graph C_{12} has an R-role assignment for $R = R_{2,4}$, $R = R_{2,5}$, or $R = R_{2,6}$.

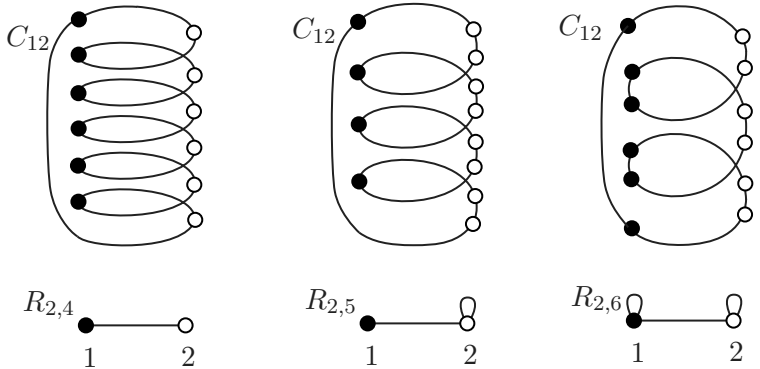


Figure 7: Possible 2-role assignments for the graph C_{12}.

State of the art

Table 1: Complexity of r-Role Assignment for certain graph classes.

Class	$r = 2$	$r = 3$	$r \geq 4$
General [Roberts and Sheng], [Fiala and Paulusma]	NP-C	NP-C	NP-C
Planar [Purcell and Rombach]	NP-C	NP-C	NP-C
Maximum degree 3 [Purcell and Rombach]	NP-C	NP-C	NP-C
Chordal [van 't Hof, Paulusma, and van Rooij]	Linear	NP-C	NP-C
Connected bipartite [Pandey]	Constant*	NP-C	NP-C
Split [Dourado]	Constant*	Linear	NP-C
K_2^2**-free** [Purcell and Rombach], [Pandey], [Dourado]	Constant	Constant*	NP-C
Unit interval [Heggernes, van 't Hof, Paulusma]	Polynomial	Polynomial	Polynomial
Cographs [Purcell and Rombach]	Constant*	Constant*	Constant*
Cartesian Product [Fernanda Mesquita's PhD]	Constant*	NP-C	NP-C
Complementary Prism [Fernanda Mesquita's PhD]	Constant	Linear	?
Maximum degree 2 [Castonguay and Nascimento]	Constant	Linear	Linear

* trivial with a positive answer.

Work in progress

Table 2: Work in progress on r-role assignment. (actualized in oct. 2024)

Product	**Factor Conditions**	$r = 2$	$r = 3$
Direct [Undergraduate Research: Luiz Macedo]	2 bipartite graphs	YES	YES
Direct [Undergraduate Research: Luiz Macedo]	1 has 3-assignment		YES
Strong [Undergraduate Research: Gustavo Medeiros]	2 general graphs	YES	
Strong [Undergraduate Research: Gustavo Medeiros]	1 is complete or bipartite	YES	YES
Corona [Undergraduate Research: Jarlilson Guajajara]	2 general graphs	YES	

Vertex coloring vs. Role assignment

Definition
A graph G admits a k-vertex coloring if there exists a mapping $c\colon V(G) \to \{1, \ldots, k\}$ such that $c(u) \neq c(v)$ for every edge $uv \in E(G)$.

When is an assignment a vertex coloring?
An R-role assignment is a vertex coloring if and only if R has no loops.

Vertex coloring vs. Role assignment

In general, a vertex coloring is not a role assignment

Example of a 3-vertex coloring of P_4 that is not a 3-role assignment.
P_4 does not even have a 3-role assignment.

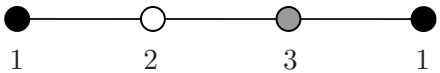


Figure 8: Graph P_4.

Vertex coloring vs. Role assignment

Continuity
The graph P_4 has an r-role assignment for $r = 1, 2, 4$, but not for $r = 3$.

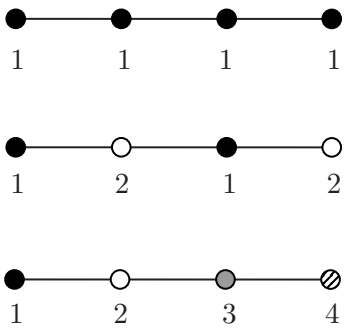


Figure 9: r-role assignments for the graph P_4, $r = 1, 2, 4$.

Vertex coloring vs. Role assignment

Connectivity

Example: C_3 and P_3 have a 3-role assignment, and $C_3 \cup P_3$ also has a 3-role assignment.

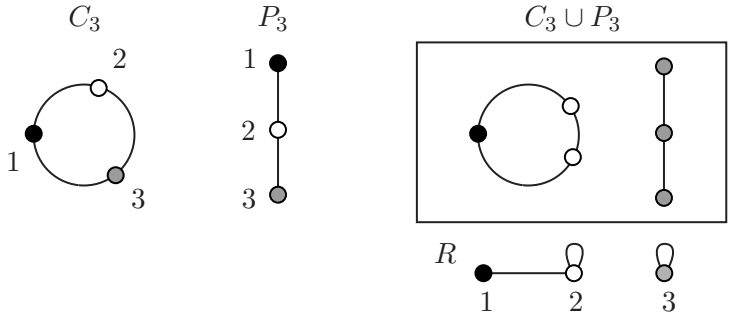


Figure 10: Distinct r-role assignments for C_3, P_3, and $C_3 \cup P_3$.

Vertex coloring vs. Role assignment

Connectivity

Example: P_{10} and C_7 have a 4-role assignment, but $P_{10} \cup C_7$ does not have a 4-role assignment.

Vertex Coloring vs. Role Assignment

Theorem 4 (-, 2024)
Let G and H be connected graphs. The graph $G \cup H$ has an r-role assignment if and only if

(i) G has an $(r-s)$-role assignment and H has an s-role assignment, or

(ii) G has an R-role assignment and H has an R-role assignment, where R is of order r.

Future Work

Table 3: Ongoing work on r-role assignment.

Class	Student
Direct Product	Undergraduate Research (IC) – Luiz Felipe Belisário Macedo
Strong Product	Undergraduate Research (IC) – Gustavo Morais Medeiros
Corona Product	Undergraduate Research (IC) – Jarlilson Guajajara Santos
Triangle-free Graphs	Undergraduate Research (IC) – Juliano Kenzo Watanabe Santana
Circulant Graphs	Undergraduate Thesis (PFC) – Maria Heloisa de Paula Souza

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Obrigada! Thank you! Merci!

Perguntas? Questions?
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